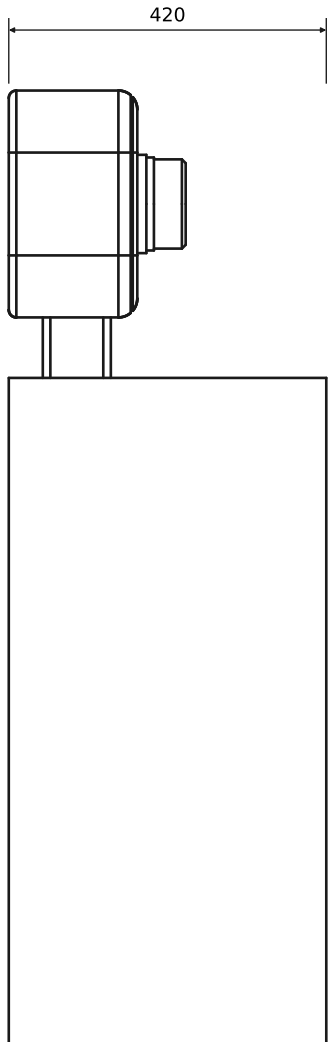
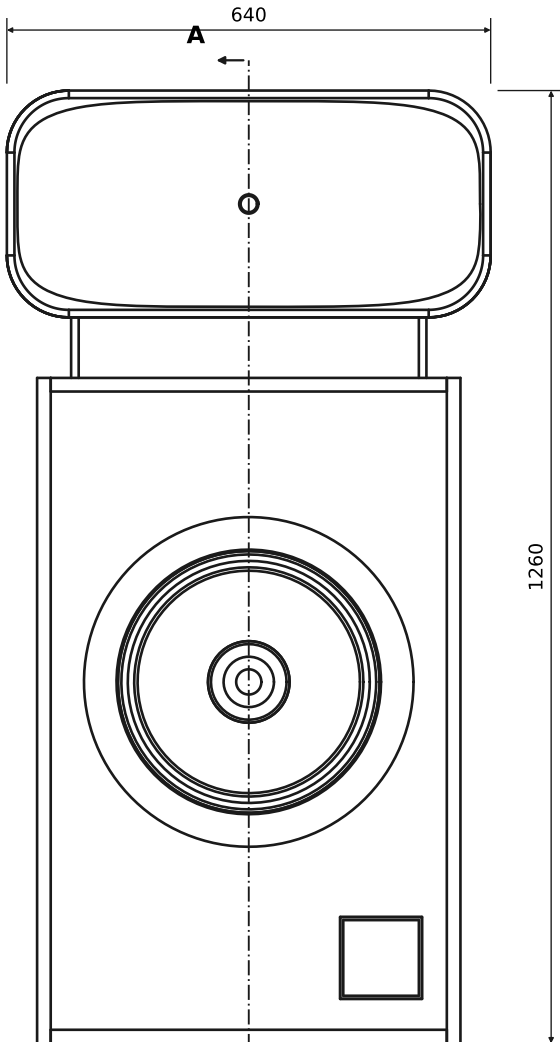


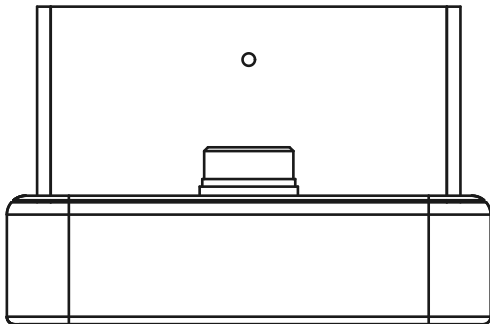
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



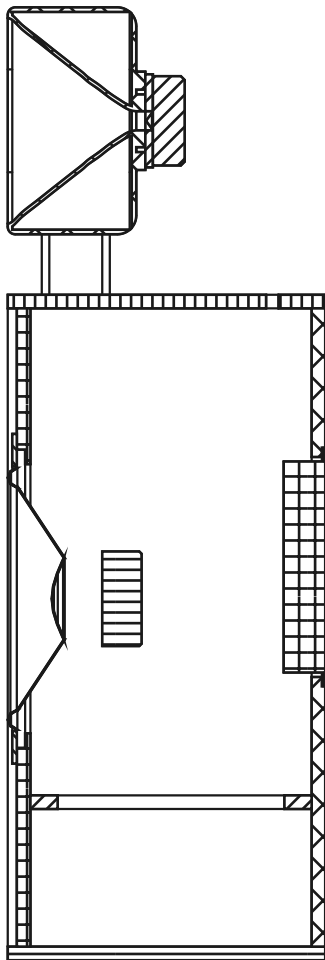
FROM ABOVE

General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.

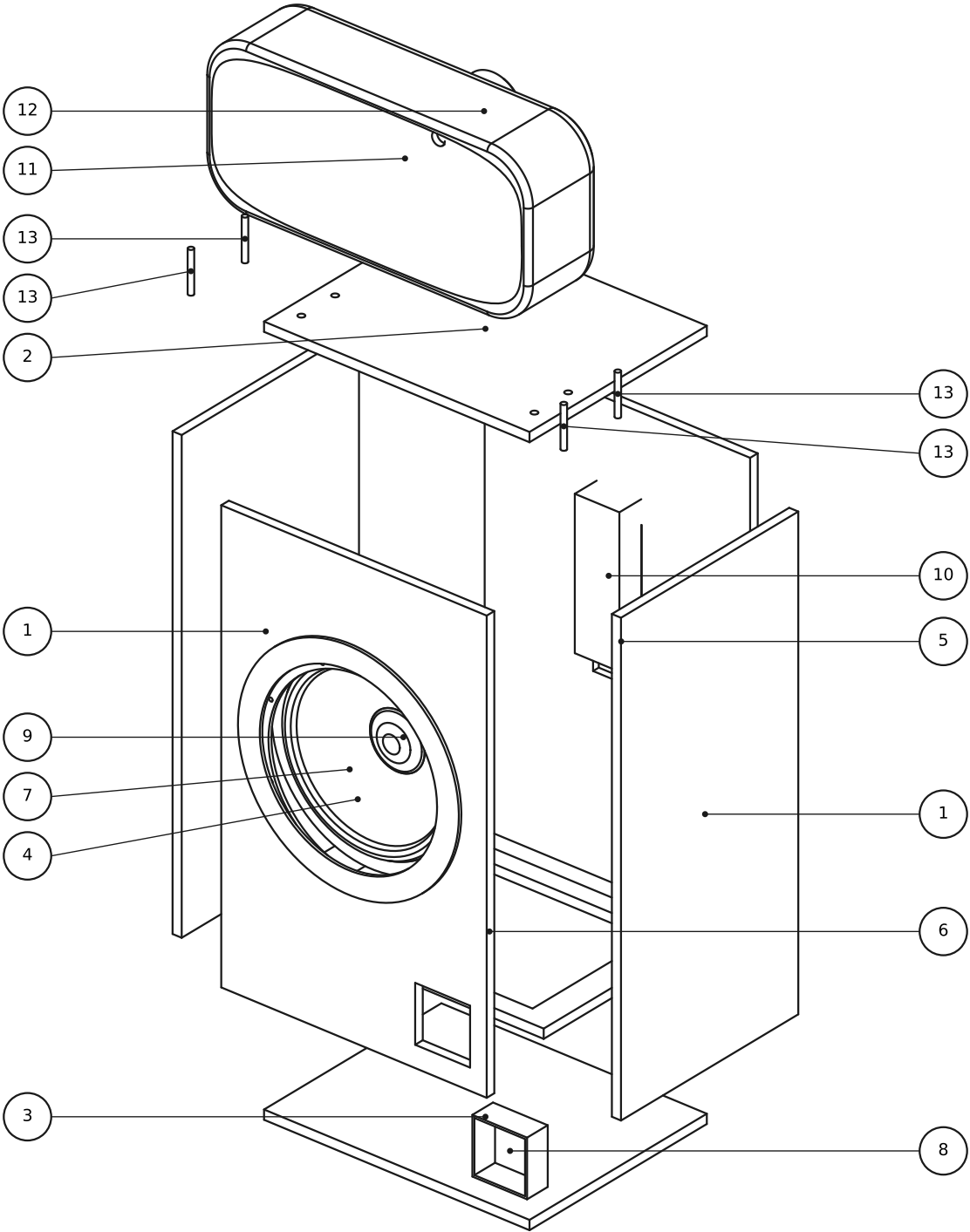


Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).

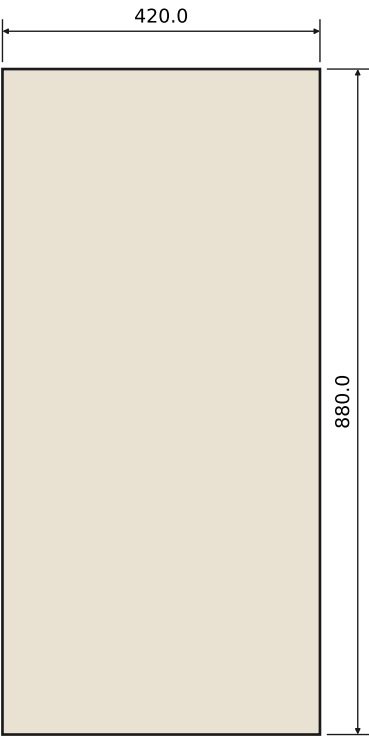


Exploded view

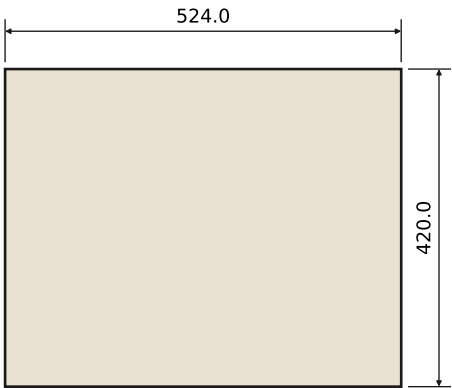
barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



left, right
18 mm baltic-birch, 4 for the set



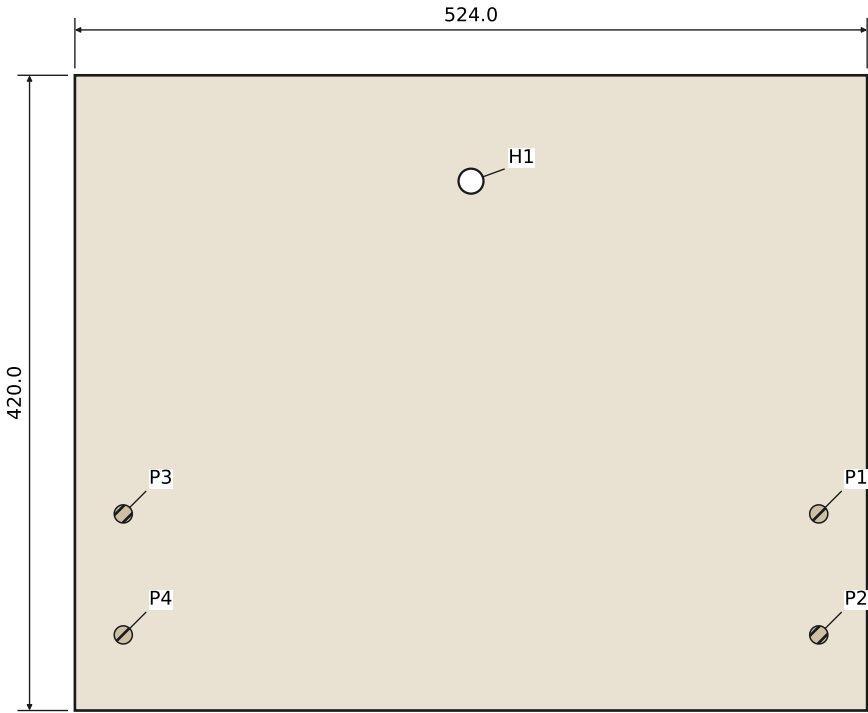
bottom
18 mm baltic-birch, 2 for the set

Sheet parts: plain panels	
barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark	
mm issue 2026-10-10 first angle Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15	4 of 12 BAR-004 rev A

Scale 1:5. As the router sees it from the outer face; DXF:
dxf/top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø12.0	492.0	130.0	15 from face
P2	Ø12.0	492.0	50.0	15 from face
P3	Ø12.0	32.0	130.0	15 from face
P4	Ø12.0	32.0	50.0	15 from face
H1	Ø16.5	262.0	350.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



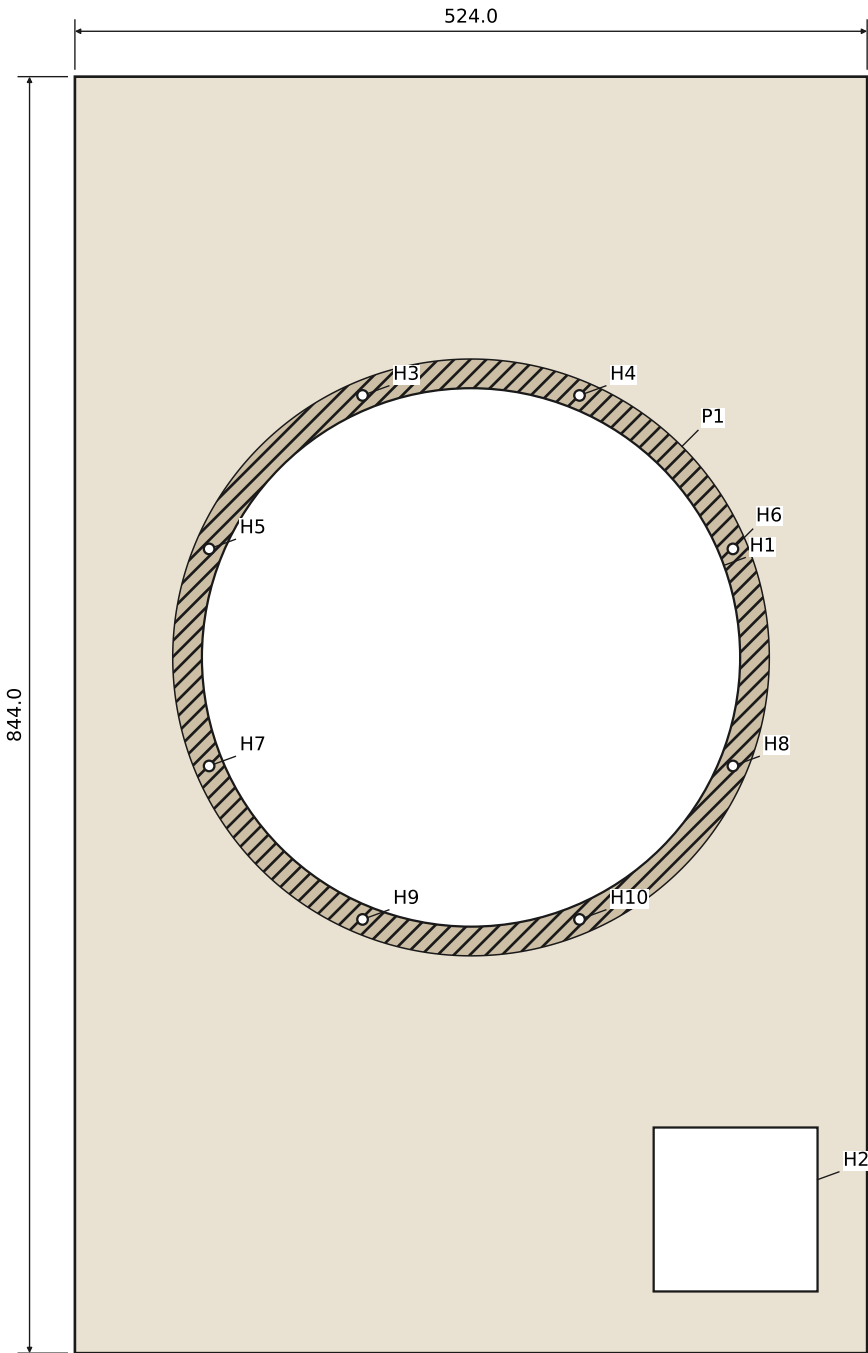
top: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 18 mm baltic-birch, 2 for the set, seen from its outer face

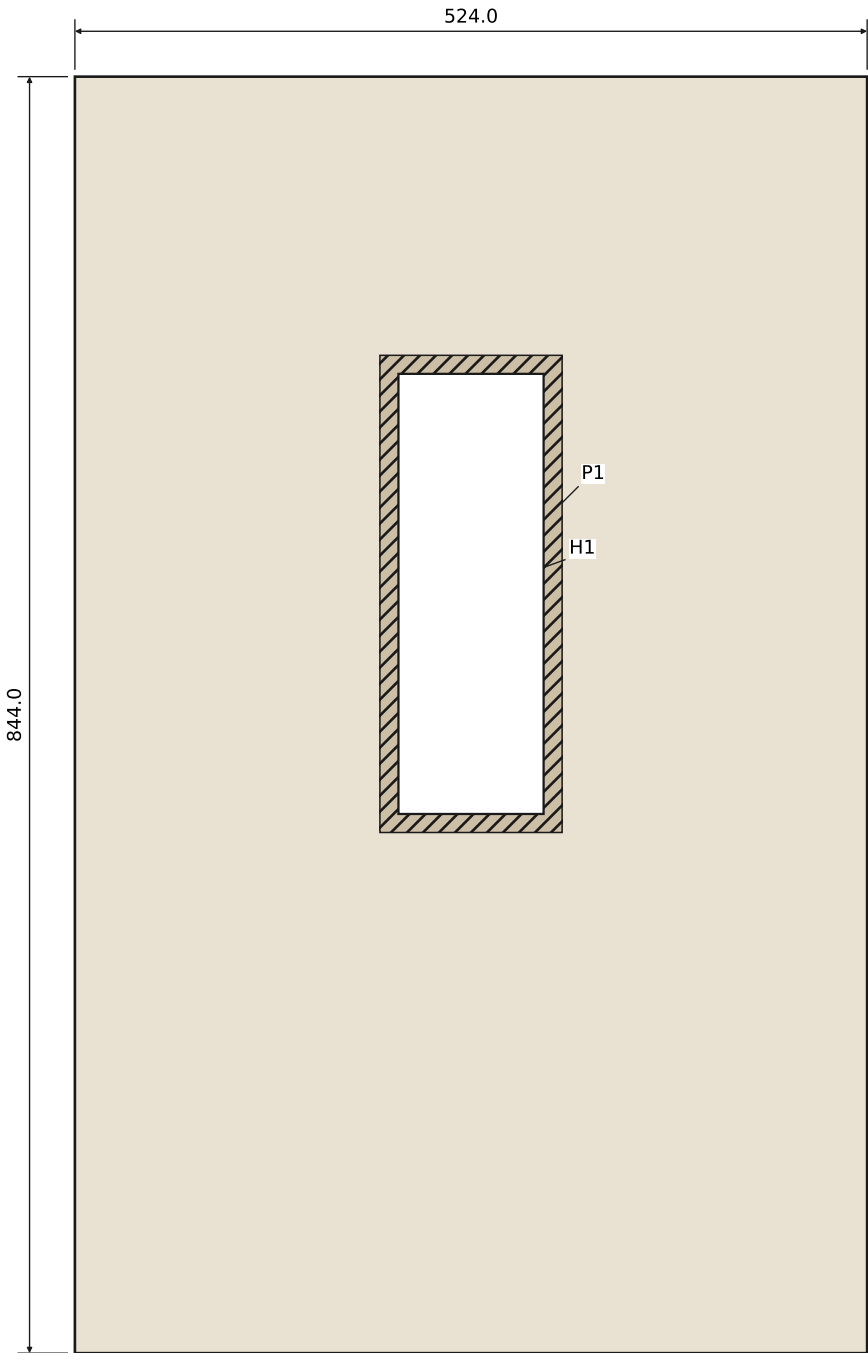
mark	size	x	y	depth
P1	Ø394.6	262.0	460.0	11 from face
H1	Ø356.0	262.0	460.0	through
H2	108.4 x 108.4	437.0	95.0	through
H3	Ø7.0	190.2	633.2	through
H4	Ø7.0	333.8	633.2	through
H5	Ø7.0	88.8	531.8	through
H6	Ø7.0	435.2	531.8	through
H7	Ø7.0	88.8	388.2	through
H8	Ø7.0	435.2	388.2	through
H9	Ø7.0	190.2	286.8	through
H10	Ø7.0	333.8	286.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	262.0	502.0	4.5 from face
H1	96.0 x 291.0	262.0	502.0	through

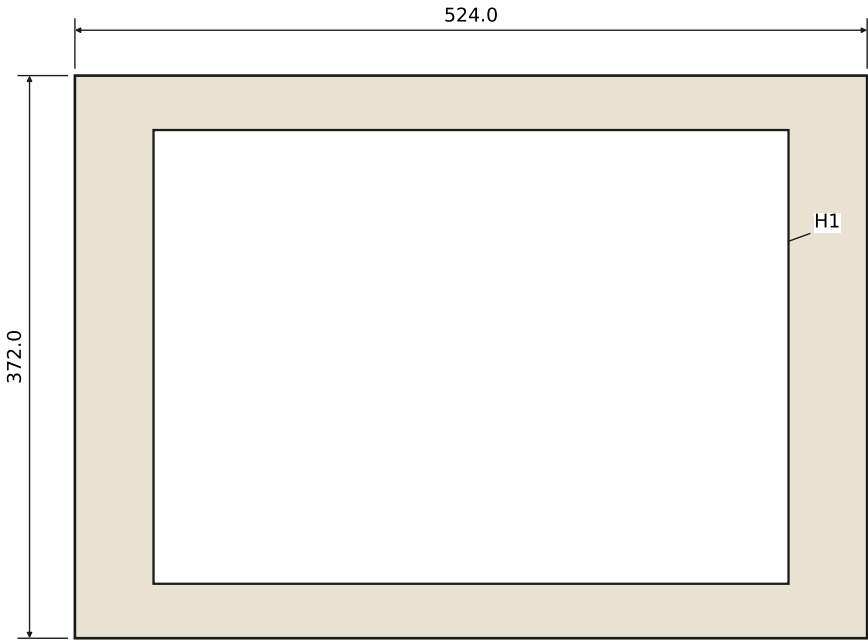
H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

Scale 1:5. As the router sees it from the outer face; DXF:
dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 300.0	262.0	186.0	through
H: cut through. P: pocket from the face shown.				
U: pocket from the inner face (dashed; its own DXF).				
Centres from the lower-left corner,				
seen from the outer face.				



brace: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: brace

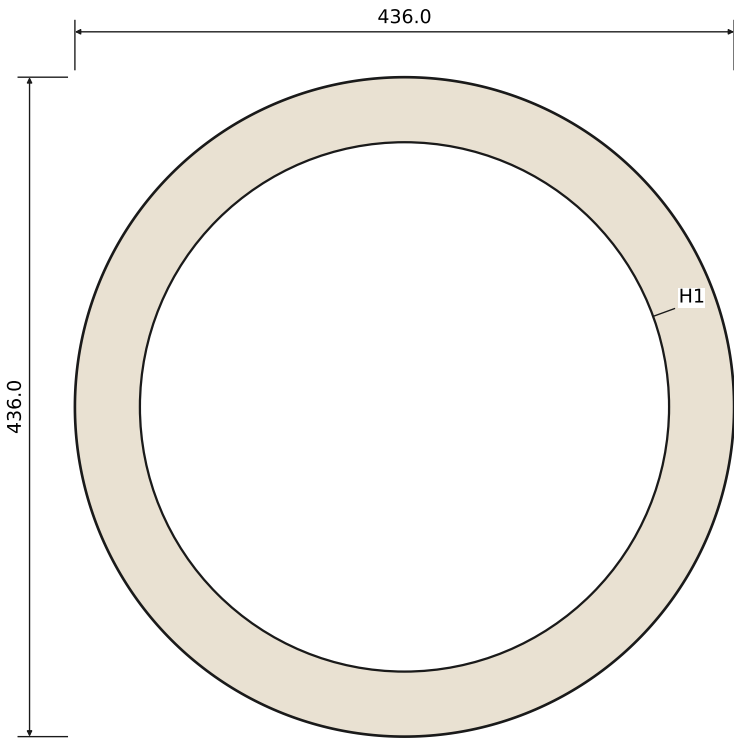
barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF:
dxf/woofer-ring.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



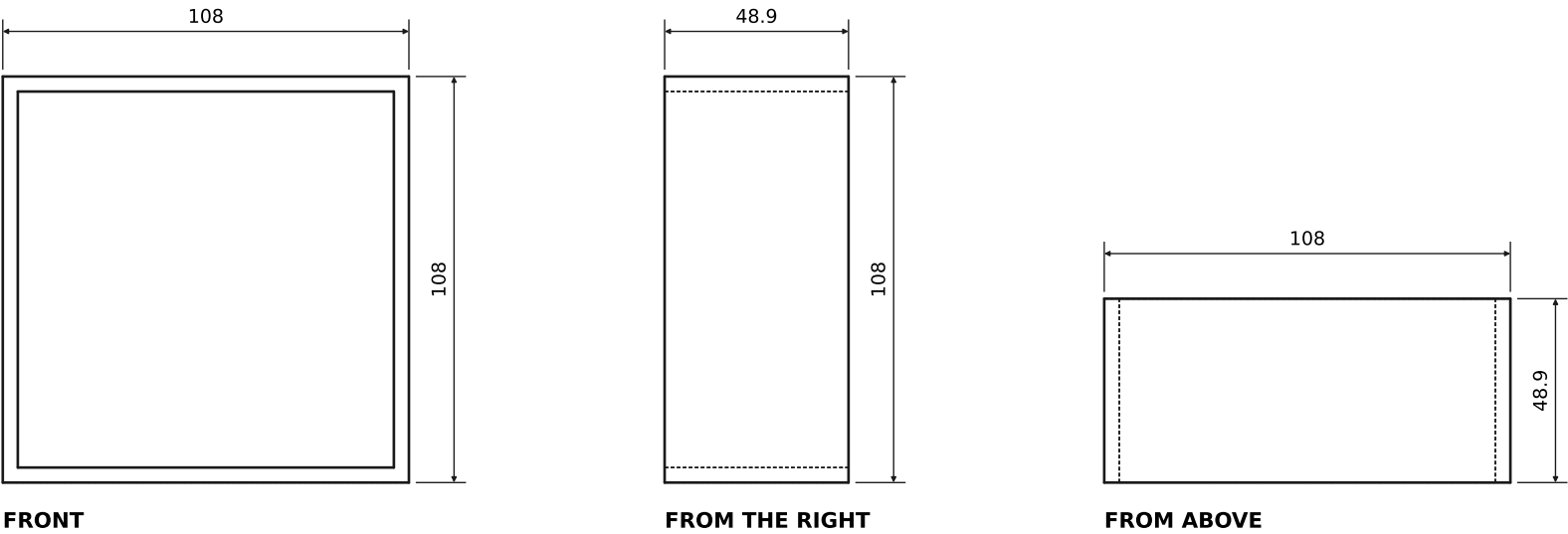
woofer-ring: 6 mm mdf, 2 for the set, seen from its outer face

Sheet part: woofer-ring

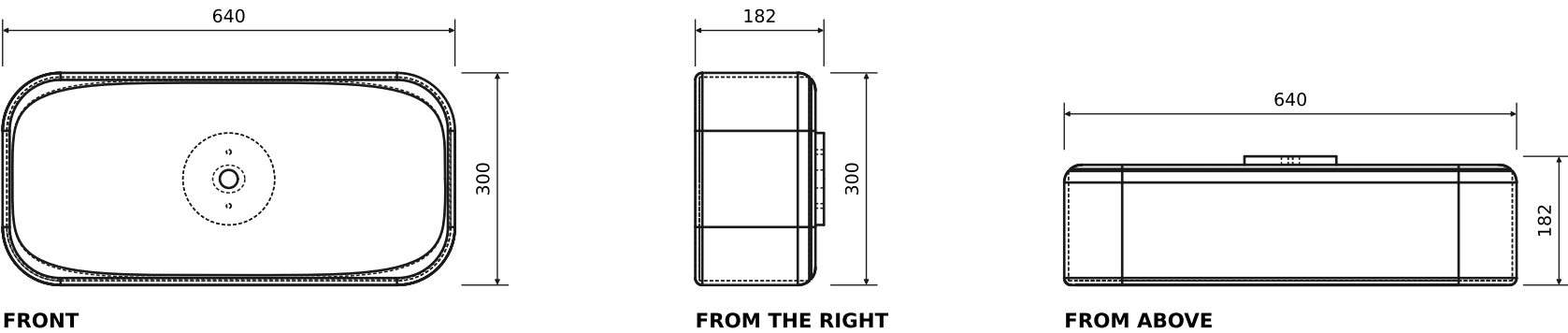
barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. port: asa print, on end; 81 cm3 each; 2 for the set; 1:2
100 square, 49 long from the panel's face: tunes the box to 38 Hz; print 15 longer and trim to tune



11. waveguide: cast, cast-resin; 3614 cm3 each; 2 for the set; 1:10
640 x 300 x 170, mouth 612 x 272, 1 in throat; cast hollow, 6 walls, in tinted translucent polyurethane in a two-part silicone mould from a CNC-cut pl



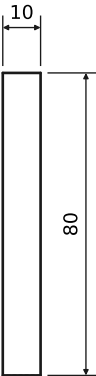
Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

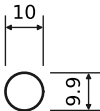
13. rod-1, rod-2, rod-3, rod-4: lathe, stainless; 6 cm3 each; 8 for the set; 1:2
M8 thread 15 deep each end



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	420 x 880	26.80
2	top	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
3	bottom	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
4	front-panel	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
5	back	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
6	brace	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 372	14.13
7	woofer-ring	cut	2	6 mm MDF	436 x 436	4.28
8	port	print	2	ASA, FDM (outdoor, paintable)	81 cm3	13.70
9	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
10	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
11	waveguide	cast	2	Tinted polyurethane casting resin, translucent	3614 cm3, 650 x 192 x 310	614.03
12	hf	buy	2	BMS 4550-16 compression driver, 1 in exit, polyester diaphragm, 16 ohm	123 across, 52 deep, 2 x M6 on 76.2	154.00
13	rod-1, rod-2, rod-3, rod-4	lathe	8	Stainless steel 304	20 x 20 x 90	11.08
14	woofer-fixings	buy	4	M4 x 20 low-profile button heads (A2) with M4 T-nuts, a driver's set of 4	M4	4.00
15	hf-fixings	buy	2	M6 x 25 socket caps with washers (the compression driver to the horn's flange), a pair		[PRICE]
16	gasket	buy	4	1.5 mm closed-cell foam gasket strip, 1 m		5.00
17	damping	buy	4	Polyester fibre fill, 150 g		8.00
18	amp-fixings	buy	2	Hypex 4.3 x 16 self-tapping pan heads and 3 mm EPDM tape, a plate's set		10.00
19	cable	buy	6	Round-sheathed speaker cable 2 x 1.5 mm2, 1 m		2.00
20	terminals	buy	8	6.3 mm insulated push-on terminal		0.50
21	hf-pad	buy	2	Resistor L-pad for the HF channel, about -10 dB at 16 ohm (lowers the amplifier's hiss through a 113 dB driver)		[PRICE]
22	glides	buy	2	Felt glides 30 mm, screw-in, a set of 4		[PRICE]

Notes

- 1. Bark on the shell (sides, top, bottom and back): real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag and sealed with a matt waterborne lacquer, wrapped round the shell's front edges; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
- 2. Glue-up: sides to the bottom, the brace, then the front panel (set back 12 mm), the back, then the top.
- 3. The woofer flush in its rebate on a gasket, eight M6 T-nuts from inside, the ring over its frame; the port pushed in dry, trimmed to tune (impedance minimum at fb), then bonded.
- 4. The waveguide stands on four 10 mm stainless rods, M8 at both ends, in inserts in the top and in bosses cast inside its underside.
- 5. The FA122 crosses at about 800 Hz (LR4) with the HF channel about 14 dB down, part of it in the passive L-pad after the amplifier, so its hiss stays inaudible through a 113 dB driver.
- 6. Hypex advises a sealed box round the amplifier module (left out here, as in the kit's example).

Parts list and notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15